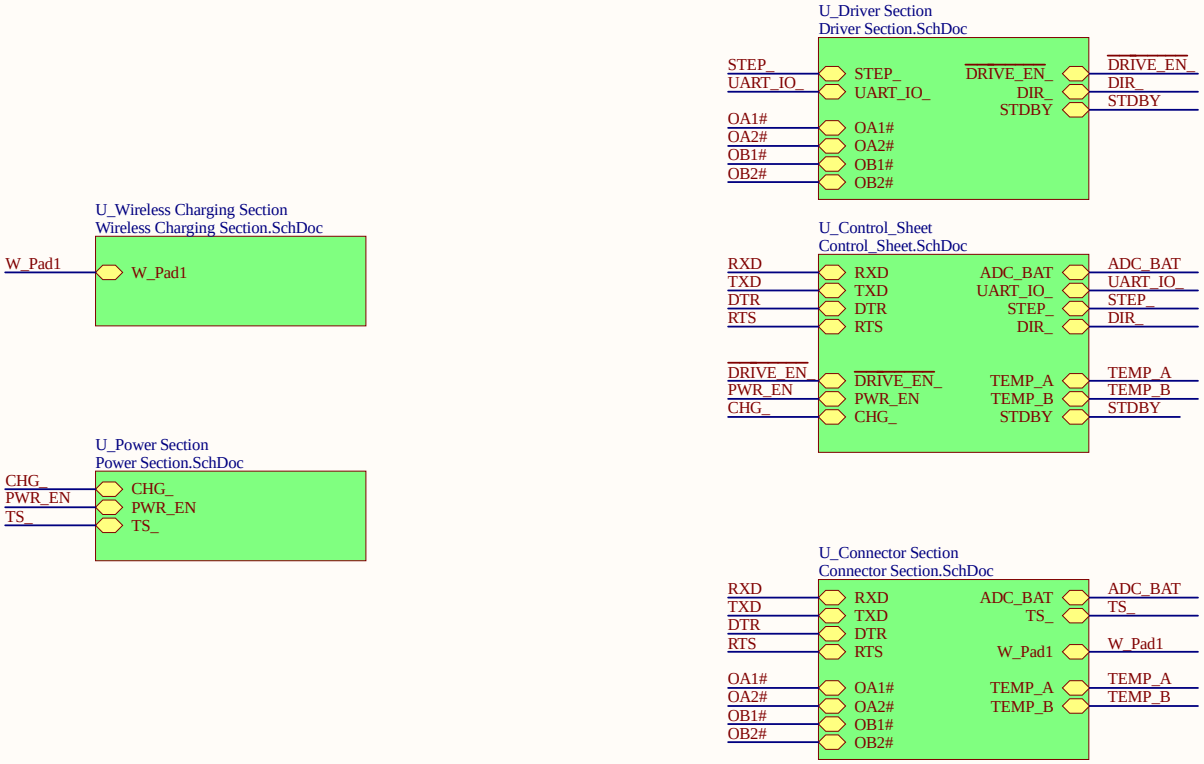
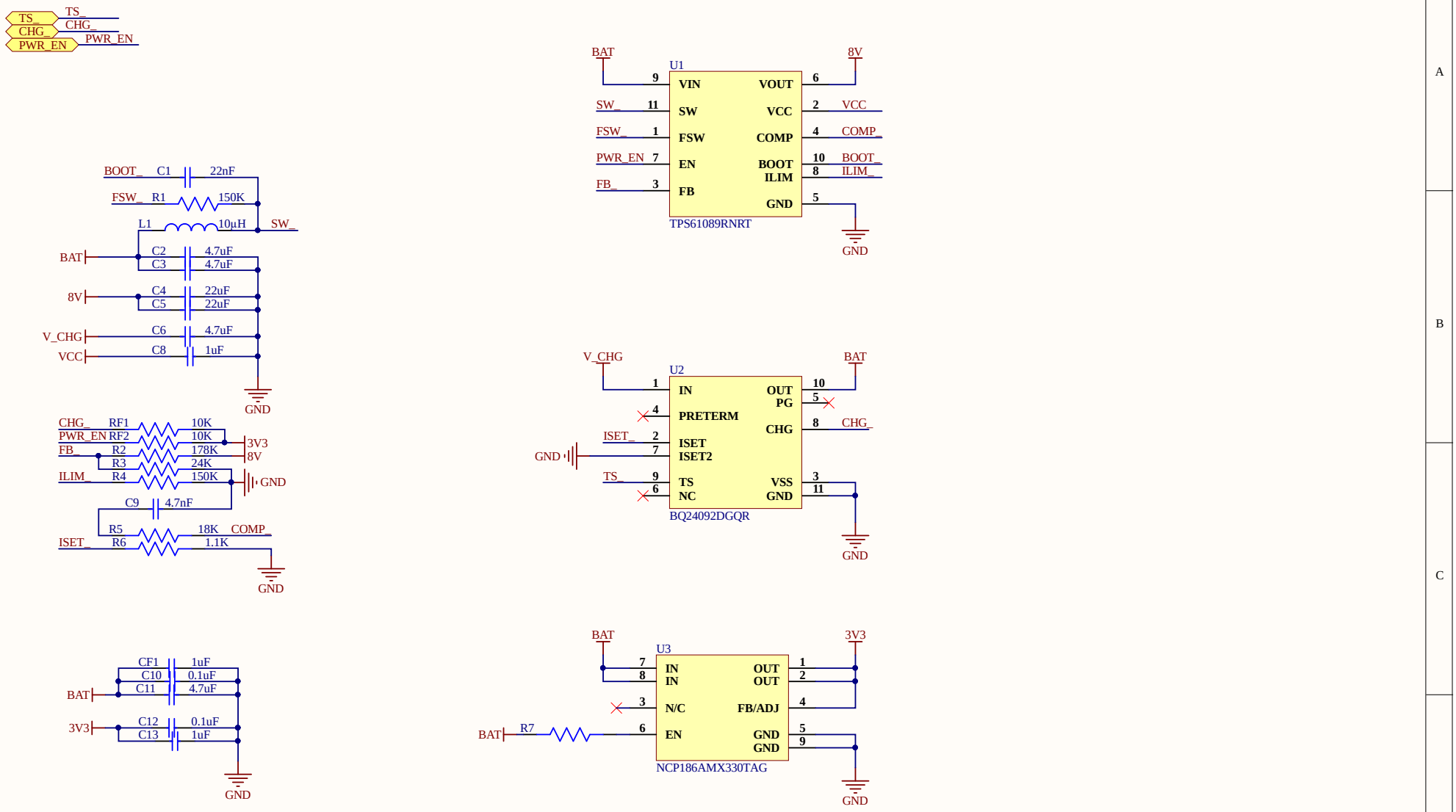
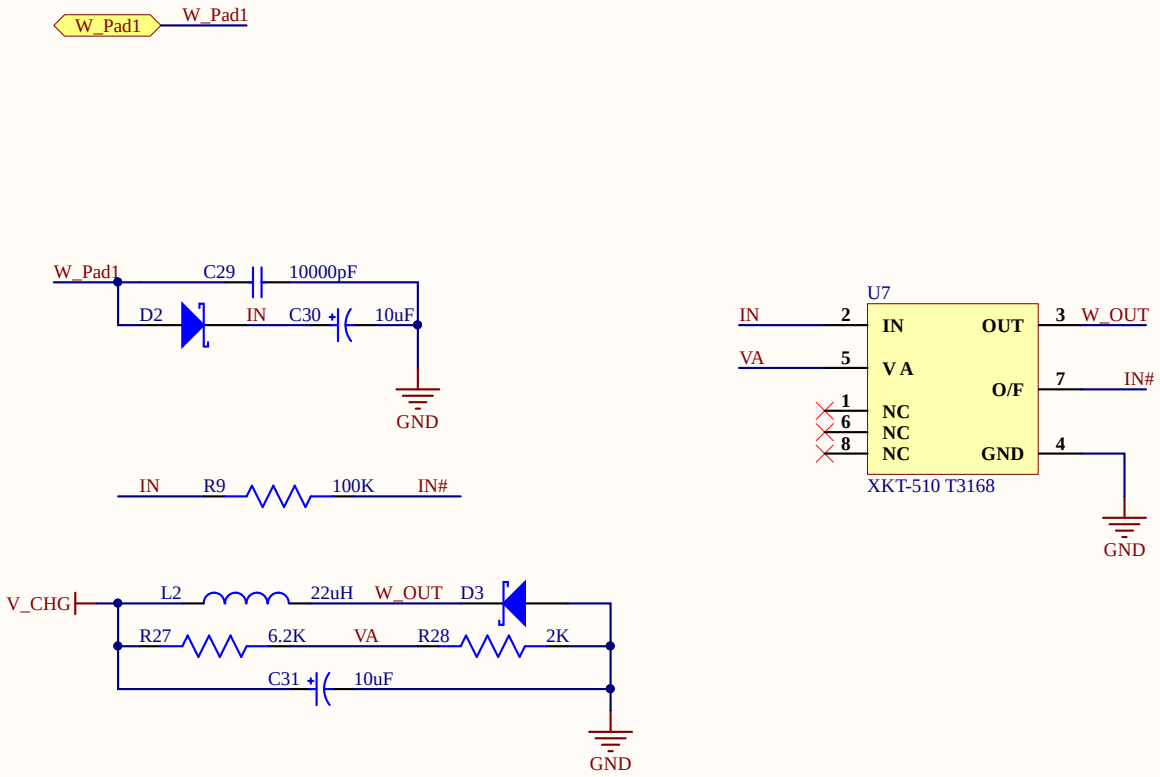
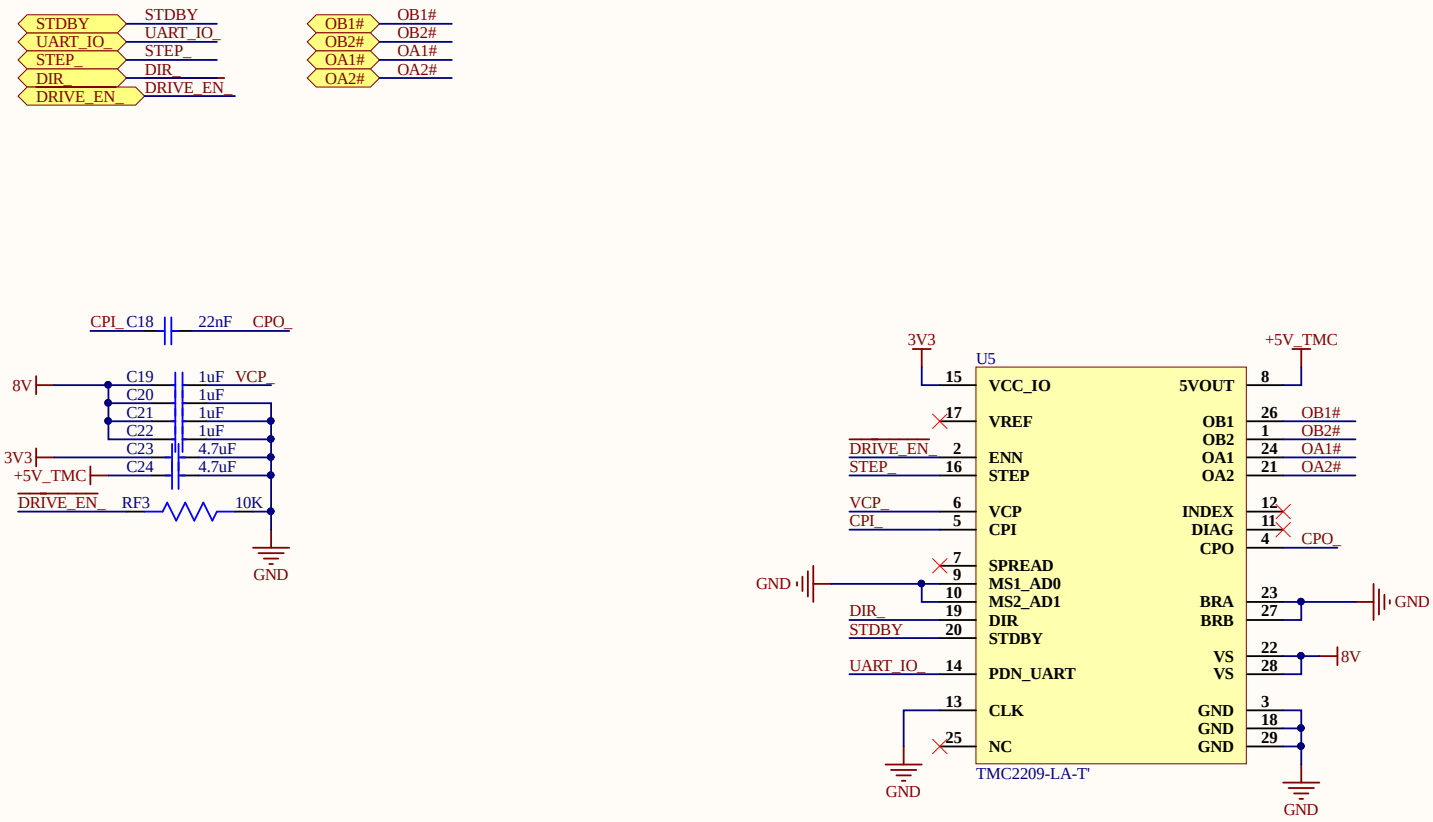
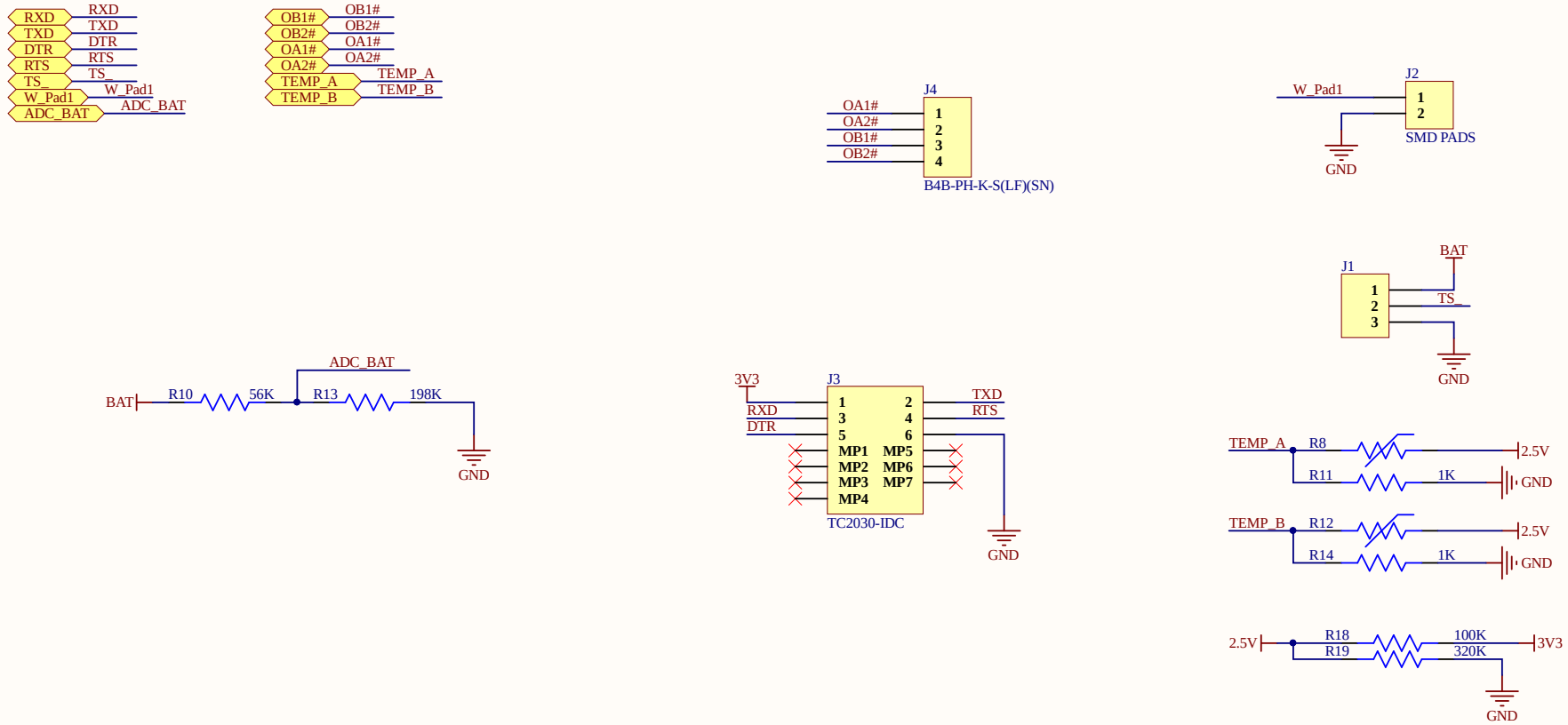

U_Wireless Charging Section
Wireless Charging Section.SchDoc
W_Pad1
W_Pad1
U_Power Section
Power Section.SchDoc
CHG_
PWR_EN
TS_
CHG_
PWR_EN
TS_
U_Driver Section
Driver Section.SchDoc
STEP_
UART_IO_
OA1#
OA2#
OB1#
OB2#
STEP_
UART_IO_
OA1#
OA2#
OB1#
OB2#
DRIVE_EN_
DIR_
STDBY
DRIVE_EN_
DIR_
STDBY
U_Control_Sheet
Control_Sheet.SchDoc
RXD
TXD
DTR
RTS
DRIVE_EN_
PWR_EN
CHG_
RXD
TXD
DTR
RTS
DRIVE_EN_
PWR_EN
CHG_
ADC_BAT
UART_IO_
STEP_
DIR_
TEMP_A
TEMP_B
STDBY
ADC_BAT
UART_IO_
STEP_
DIR_
TEMP_A
TEMP_B
STDBY
U_Connector Section
Connector Section.SchDoc
RXD
TXD
DTR
RTS
OA1#
OA2#
OB1#
OB2#
RXD
TXD
DTR
RTS
OA1#
OA2#
OB1#
OB2#
ADC_BAT
TS_
W_Pad1
TEMP_A
TEMP_B
ADC_BAT
TS_
W_Pad1
TEMP_A
TEMP_B

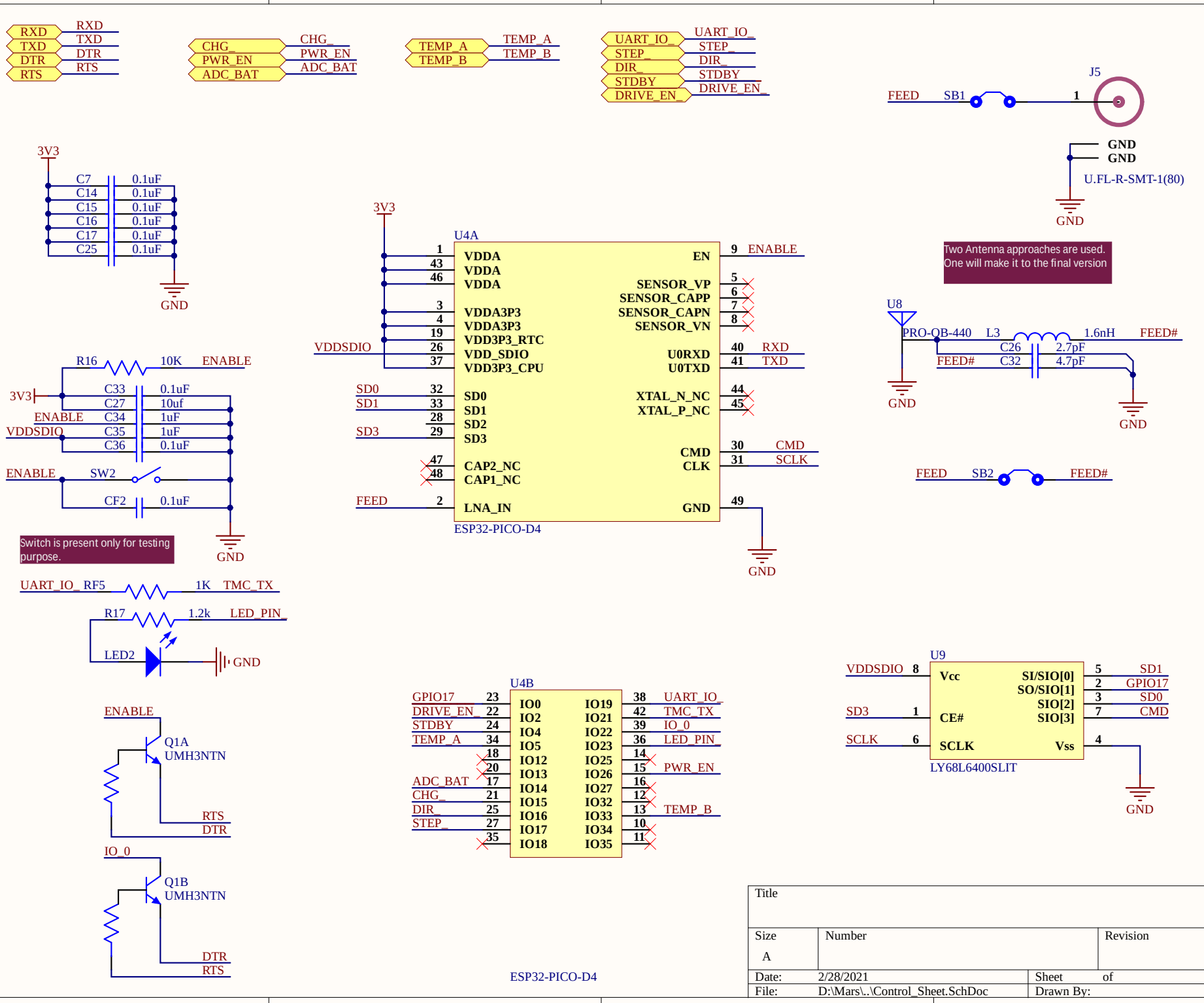

TS_
CHG_
PWR_EN
BOOT_ C1 22nF
FSW_ R1 150K
L1 10µH SW_
BAT C2 4.7uF
C3 4.7uF
8V C4 22uF
C5 22uF
V_CHG C6 4.7uF
VCC C8 1uF
GND
CHG_ RF1 10K
PWR_EN RF2 10K
FB_ R2 178K
R3 24K
ILIM_ R4 150K
3V3
8V
GND
C9 4.7nF
R5 18K COMP_
ISET_ R6 1.1K
GND
CF1 1uF
C10 0.1uF
BAT C11 4.7uF
3V3 C12 0.1uF
C13 1uF
GND
BAT
U1
9 VIN
11 SW
1 FSW
PWR_EN 7 EN
FB_ 3 FB
VOUT 6
VCC 2
COMP 4
BOOT 10
ILIM 8
GND 5
8V
VCC
COMP_
BOOT_
ILIM_
GND
TPS61089RNRT
V_CHG
U2
1 IN
4 PRETERM
ISET_ 2
GND 7 ISET
ISET2
TS_ 9 TS
6 NC
OUT 10
PG 5
CHG 8
VSS 3
GND 11
BAT
CHG_
GND
BQ24092DGQR
BAT
U3
7 IN
8 IN
3 N/C
BAT R7 6 EN
OUT 1
OUT 2
FB/ADJ 4
GND 5
GND 9
3V3
GND
NCP186AMX330TAG



Title		
Size: A	Number	Revision
Date: 2/28/2021	Sheet of	
File: D:\Mars\..\Wireless Charging Section.SchDoc	Drawn By:	


STDBY
UART_IO_
STEP_
DIR_
DRIVE_EN_
OB1#
OB2#
OA1#
OA2#
CPI_C18 22nF CPO_
8V
C19 1uF VCP_
C20 1uF
C21 1uF
C22 1uF
3V3
C23 4.7uF
+5V_TMC
C24 4.7uF
DRIVE_EN_ RF3 10K
GND
U5
3V3
+5V_TMC
15 VCC_IO
5VOUT 8
17 VREF
OB1 26 OB1#
OB2 1 OB2#
DRIVE_EN_ 2 ENN
OA1 24 OA1#
STEP_ 16 STEP
OA2 21 OA2#
VCP_ 6 VCP
INDEX 12
CPI_ 5 CPI
DIAG 11
CPO 4 CPO_
7 SPREAD
GND 9 MS1_AD0
10 MS2_AD1
BRA 23
DIR_ 19 DIR
BRB 27
STDBY 20 STDBY
VS 22
8V
UART_IO_ 14 PDN_UART
VS 28
13 CLK
GND 3
GND 18
25 NC
GND 29
TMC2209-LA-T'
GND
GND
GND


RXD
TXD
DTR
RTS
TS_
W_Pad1
ADC_BAT
OB1#
OB2#
OA1#
OA2#
TEMP_A
TEMP_B
J4
B4B-PH-K-S(LF)(SN)
J2
SMD PADS
GND
J1
BAT
3V3
J3
TC2030-IDC
MP1
MP2
MP3
MP4
MP5
MP6
MP7
R10
56K
R13
198K
R8
R11
1K
R12
R14
2.5V
R18
100K
R19
320K


RXD
TXD
DTR
RTS
CHG_
PWR_EN
ADC_BAT
TEMP_A
TEMP_B
UART_IO_
STEP_
DIR_
STDBY
DRIVE_EN_
J5
FEED
SB1
GND
U.FL-R-SMT-1(80)
3V3
C7 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 0.1uF
C25 0.1uF
U4A
ESP32-PICO-D4
VDDSDIO
SD0
SD1
SD3
CAP2_NC
CAP1_NC
LNA_IN
EN
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
U0RXD
U0TXD
XTAL_N_NC
XTAL_P_NC
CMD
CLK
Two Antenna approaches are used.
One will make it to the final version
U8
PRO-QB-440
L3
1.6nH
FEED#
C26 2.7pF
C32 4.7pF
SB2
R16 10K ENABLE
C33 0.1uF
C27 10uf
C34 1uF
C35 1uF
C36 0.1uF
SW2
CF2 0.1uF
Switch is present only for testing purpose.
R17 1.2k LED_PIN_
LED2
UART_IO_ RF5 1K TMC_TX
U4B
GPIO17
DRIVE_EN_
IO_0
PWR_EN
TMC_TX
LED_PIN_
U9
LY68L6400SLIT
Vcc
CE#
SCLK
Vss
SI/SIO[0]
SO/SIO[1]
SIO[2]
SIO[3]
Q1A
UMH3NTN
Q1B
UMH3NTN
Title
Size
A
Number
Revision
Date: 2/28/2021
Sheet of
File: D:\Mars\..\Control_Sheet.SchDoc
Drawn By: